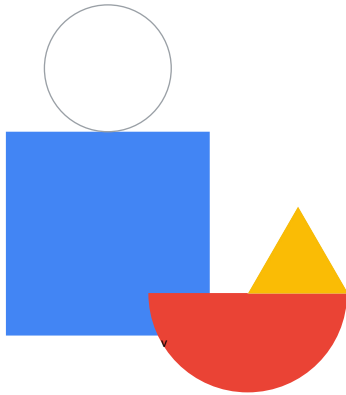


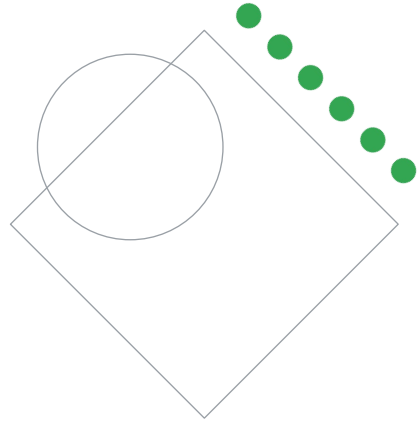


Preparing for Your Professional Cloud Network Engineer Journey

Module 3: Configuring managed network services

Welcome to Module 3: Configuring managed network services.

Review and study planning

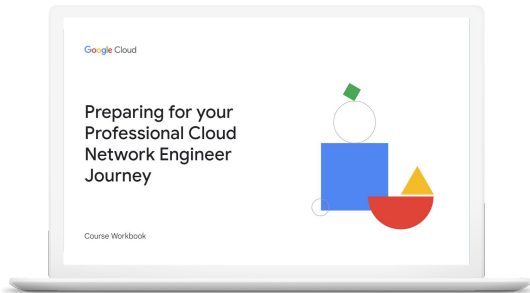


Google Cloud

You'll now review the diagnostic questions and your answers to help you identify what to include in your study plan.

Your study plan:

Configuring managed network services



3.1

Configuring load balancing

3.2

Configuring Cloud CDN

3.3

Configuring Cloud DNS

Google Cloud

The diagnostic questions align with these objectives of this exam section. Use the PDF resource that follows to review the questions and how you answered them. Pay specific attention to the rationale for both the correct and incorrect answers. Use the resources detailed under **Where to look** and **Content mapping** to build a study plan that meets your learning needs.

3.1 | Configuring load balancing (1)

Considerations include:

- Determining the load balancing solution for your network (Internal/External, Regional/Global, Application/Proxy/Passthrough, etc)
- Configuring backend services (e.g., network endpoint groups (NEGs), managed instance groups)
- Configuring various load balancers and backend settings such as the balancing method, session affinity, serving capacity, URL maps, health checks, and global access
- Optimizing for traffic scalability by using autoscaling or manual scaling features

Google Cloud

A Professional Cloud Network Engineer should be familiar with load balancing, coupled with autoscaling, as a method of provide scalability, resilience, availability, and performance. You should also be familiar with the varieties available in Google Cloud, their tradeoffs, and the details involved in their implementation.

Question 1 tested your knowledge of load balancer types and their configuration details. Question 2 tested your knowledge of managed instance groups and autoscaling. Question 3 and Question 4 tested your networking knowledge of the OSI model, and which load balancing options are appropriate for Layer 4 and Layer 7 traffic.

To answer these questions, you should be comfortable with all of the considerations listed on this slide and the next.

3.1 | Configuring load balancing (2)

Considerations include:

- Understanding load balancers in GKE (e.g., GKE Gateway controller, GKE Ingress controller, NEG)
- Setting up traffic management on Application Load Balancers (e.g., traffic splitting, traffic mirroring, URL rewrites)

3.1 Diagnostic Question 01 Discussion



Cymbal Bank wants a web application to have global anycast load balancing across multiple regions. The web application will serve static asset files and will also use REST APIs that serve dynamic responses. The load balancer should support HTTP and HTTPS requests and redirect HTTP to HTTPS. The load balancer should also serve all the requests from the same domain name, with different paths indicating static versus dynamic resources.

Select the load balancer configuration that would most effectively enable this scenario.

- A. A global external Application Load Balancer with one global forwarding rule, forwarding to one target proxy with one URL map connected to two backend services
- B. A global external Application Load Balancer with two global forwarding rules, forwarding to two target proxies: one with URL map and no backend service, and the other with URL map and two backend services
- C. Two global external Application Load Balancers, each with one global forwarding rule forwarding to one target proxy with one URL map connected to one backend service
- D. **A global external Application Load Balancer with two global forwarding rules forwarding to two target proxies: one with URL map and no backend service, and the other with URL map, one backend service, and one backend bucket**

Google Cloud

Feedback:

A. Incorrect. A single target proxy could not listen for both HTTP and HTTPS traffic.

B. Incorrect. Backend bucket is a more effective way (simpler, lower cost, higher availability) of serving static resources than a backend service, which must be connected to compute resources such as managed instance groups or network endpoint groups.

C. Incorrect. This solution would use two IP addresses, which would require either two domain names or DNS based load-balancing. Also, a backend bucket is a more effective way (simpler, lower cost, higher availability) of serving static resources than a backend service, which must be connected to compute resources like managed instance groups or network endpoint groups

*D. Correct! This is the most effective solution. A single global external Application Load Balancer (with a single IP address and domain name) is connected via two global forwarding rules to two target proxies. The first global forwarding rule forwarding to the first target proxy (with URL map issuing a redirect, and no connected backend service) serves HTTP requests and simply redirects to HTTPS. The second global forwarding rule forwards to the second target proxy. This serves HTTPS and uses the URL map to forward static resource requests to either a backend bucket or a backend service, based on the path of the request. The backend bucket serves the web-application files and connects to a Cloud Storage bucket. The backend service serves dynamic requests, and is connected to a managed instance group or network endpoint group to serve those dynamic requests via a REST API.

Where to look:

<https://cloud.google.com/load-balancing/docs/load-balancing-overview>
<https://cloud.google.com/load-balancing/docs/choosing-load-balancer>
<https://cloud.google.com/load-balancing/docs/features>
<https://cloud.google.com/load-balancing/docs/https>
<https://cloud.google.com/load-balancing/docs/l7-internal>
<https://cloud.google.com/load-balancing/docs/network/networklb-backend-service>
<https://cloud.google.com/load-balancing/docs/internal>
<https://cloud.google.com/load-balancing/docs/tcp>
<https://cloud.google.com/load-balancing/docs/backend-service>
<https://cloud.google.com/load-balancing/docs/forwarding-rule-concepts>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M11 Hybrid Load Balancing and Traffic Management
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M1 Hybrid Load Balancing and Traffic Management
- Skill badge: Implementing Cloud Load Balancing for Compute Engine

Summary:

There are many types of load balancer available in Google Cloud. Some operate at layer 4 (L4) and others at layer 7 (L7). For HTTP and HTTPS traffic the L7 internal or external regional or global Application Load Balancer is typically the best choice.

When supporting both HTTP and HTTPS for the same domains, the recommended approach is to use 2 separate target proxies. Each target proxy requires a forwarding rule. One of the target proxies is used to redirect HTTP requests to HTTPS and requires no backend bucket or service. Each target proxy would also have a URL map that it uses to either redirect traffic, or direct traffic to backend services or backend buckets based on the domain or path of the request. This is referred to as content-based load balancing.

For global load balancers, each backend bucket or backend service can operate over many regions and will intelligently route to the nearest region to where the user traffic originates. Backend buckets are connected to Cloud Storage buckets. Backend buckets are the most effective way to serve static resources because those resources can be served directly from the Cloud Storage bucket without requiring any compute resources. Static resources can also easily be populated into Cloud CDN to be served with low latency worldwide.

3.1 Diagnostic Question 02 Discussion



You are designing a load balanced autoscaling frontend for Cymbal Bank. It is intended to be deployed into Google Kubernetes Engine (GKE). You want to use container-native load balancing and autoscale based on the amount of traffic to the service.

Select the type of backend and autoscaling that would accomplish this.

- A. A managed instance group of GKE nodes that autoscale using cluster autoscaling
- B. A zonal network endpoint group of Kubernetes pods that autoscale using a Horizontal Pod Autoscaler**
- C. A managed instance group of GKE nodes that contain pods that autoscale using a Horizontal Pod Autoscaler
- D. A serverless network endpoint group of GKE pods that autoscale using a Horizontal Pod Autoscaler

Google Cloud

Feedback:

A. Incorrect. Managed instance groups of VMs are not used for container-native load balancing. They are used for non container-native load balancing in GKE. Also, cluster autoscaling is based on the resource demands of all workloads in the cluster. Cluster autoscaling is not tied to specific workload, nor does it directly affect capacity or number of pod replicas for a particular workload. It has an indirect impact by allowing more or less pods to be scheduled and run.

*B. Correct! Zonal network endpoint groups connect the load balancer directly to the pods for container-native load balancing in GKE. The ReplicaSet of pod replicas can autoscale based on the amount of traffic arriving at the associated Service via the HorizontalPodAutoscaler.

C. Incorrect. Managed instance groups of the Kubernetes Engine nodes would be used for non container-native load balancing in GKE.

D. Incorrect. Serverless network endpoint groups connect load balancers to App Engine, Cloud Run, Cloud Run functions, or API Gateway services.

Where to look:

<https://cloud.google.com/compute/docs/instance-groups>

<https://cloud.google.com/compute/docs/instance-groups/creating-groups-of-managed-instances>

<https://cloud.google.com/load-balancing/docs/negs>

<https://cloud.google.com/load-balancing/docs/negs/zonal-neg-concepts>

<https://cloud.google.com/load-balancing/docs/negs/internet-neg-concepts>

<https://cloud.google.com/load-balancing/docs/negs/serverless-neg-concepts>

<https://cloud.google.com/kubernetes-engine/docs/how-to/horizontal-pod-autoscaling#autoscale-traffic>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M11 Hybrid Load Balancing and Traffic Management
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M1 Hybrid Load Balancing and Traffic Management
- Skill badge:
 - Build a Secure Google Cloud Network

Summary:

Managed instance groups and network endpoint groups can both autoscale. They create or delete replicas to serve traffic automatically, based on resource target values that are measured by metrics collected by Google Cloud Monitoring. Managed instance groups are always groups of identical VMs. Network endpoint groups can be more general, such as FQDN/port or IP/port combinations, or managed services. In GKE you can use either managed instance groups or network endpoint groups, but container-native load balancing only uses network endpoint groups.

In GKE workloads, autoscaling is typically accomplished using a HorizontalPodAutoscaler, though VerticalPodAutoscaler and MultidimPodAutoscaler are possible alternatives. GKE cluster autoscaling occurs based on the resource demands and scheduling of pods across all workloads using a given node pool.

3.1 Diagnostic Question 03 Discussion



Cymbal Bank is migrating a legacy TCP-based trading application to Google Cloud. The application runs on a specific port (not HTTP/HTTPS) and requires the backend servers to see the original client IP address of the incoming traffic for compliance auditing. The backend servers are located in a single region (us-central1).

- A. Global External Application Load Balancer
- B. Global External Proxy Network Load Balancer
- C. Regional External Passthrough Network Load Balancer**
- D. Internal Passthrough Network Load Balancer

Select the load balancing configuration that meets the objective.

Google Cloud

Feedback:

- A. Incorrect. This load balancer operates at Layer 7 (HTTP/HTTPS) and cannot handle raw TCP traffic on arbitrary ports.
- B. Incorrect. While this handles TCP, it is a proxy. The backend sees the IP of the Load Balancer, not the client, unless the proxy protocol is enabled and the app supports it (which legacy apps often do not).
- *C. Correct! The Regional External Passthrough Network Load Balancer operates at Layer 4 and preserves the original client IP address by passing packets directly to backends, which satisfies compliance requirements without additional configuration. Additionally, because the backend servers handle TCP traffic in a single region, this load balancer is the most architecturally appropriate and cost-effective choice.
- D. Incorrect. The scenario implies external clients (trading application) need access; Internal load balancers are not reachable from the public internet.

Where to look:

<https://docs.cloud.google.com/load-balancing/docs>
<https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer>
<https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer#traffic-type>
<https://docs.cloud.google.com/load-balancing/docs/application-load-balancer>
<https://docs.cloud.google.com/load-balancing/docs/proxy-network-load-balancer>
<https://docs.cloud.google.com/load-balancing/docs/passthrough-network-load-balancer>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M11 Hybrid Load Balancing and Traffic Management
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M1 Hybrid Load Balancing and Traffic Management
- Skill badges:
 - Build a Secure Google Cloud Network

Summary:

The Regional External Passthrough Network Load Balancer is a Layer 4 load balancer that preserves the client's IP address. Because it is "passthrough," the packets sent to the backend VMs have the source IP of the client, not the load balancer. This satisfies the compliance requirement without needing complex headers or PROXY protocol configurations.

Additionally, since the backend servers are located in a single region (us-central1), a regional load balancer is the most architecturally appropriate and cost-effective choice. It is designed specifically for UDP, TCP, and ESP traffic that is not HTTP-based.

3.1 Diagnostic Question 04 Discussion



Cymbal Bank is developing a new internal employee dashboard hosted on a VPC in us-east1. The application is built using microservices (e.g., /hr, /finance, /it). You need a load balancing solution that is not exposed to the public internet and can route traffic to different backend instance groups based on the URL path.

- A. Internal Passthrough Network Load Balancer
- B. Regional External Application Load Balancer
- C. Cross-region Internal Application Load Balancer
- D. **Regional Internal Application Load Balancer**

Select the load balancing solution that meets the objective.

Google Cloud

Feedback:

A. Incorrect. This is a Layer 4 load balancer. It routes based on IP and Port, not URL paths (Layer 7).

B. Incorrect. The requirement specifically states the dashboard is for internal employees and should not be exposed to the public internet.

C. Incorrect. While technically capable of path routing, the scenario specifies the app is hosted in a single region (us-east1). A Regional Internal Application Load Balancer is the standard choice unless high availability across multiple regions is explicitly required.

*D. Correct! The Regional Internal Application Load Balancer is a Layer 7 proxy-based load balancer that runs strictly within the VPC. It supports URL maps, which allow you to route traffic to different backend services (like HR or Finance microservices) based on the path (e.g., cymbal.internal/hr).

Where to look:

<https://docs.cloud.google.com/load-balancing/docs>

<https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer>

<https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer#traffic-type>

<https://docs.cloud.google.com/load-balancing/docs/application-load-balancer>

<https://docs.cloud.google.com/load-balancing/docs/proxy-network-load-balancer>

<https://docs.cloud.google.com/load-balancing/docs/passthrough-network-load-balancer>

<https://docs.cloud.google.com/load-balancing/docs/l7-internal>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M11 Hybrid Load Balancing and Traffic Management
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M1 Hybrid Load Balancing and Traffic Management
- Skill badges:
 - Build a Secure Google Cloud Network

Summary:

The Regional Internal Application Load Balancer is a Layer 7 proxy-based load balancer that runs strictly within the VPC. It supports URL maps, which allow you to route traffic to different backend services (like HR or Finance microservices) based on the path (e.g., cymbal.internal/hr).

Because the application and the intended user base are internal, this load balancer ensures the traffic never traverses the public internet, maintaining a stronger security posture. The "Regional" aspect fits the requirement that the workload is hosted specifically in us-east1.

3.1 | Configuring load balancing

Courses



[Networking in Google Cloud](#)

- M11 Hybrid Load Balancing and Traffic Management



[Networking in Google Cloud: Load Balancing](#)

- M1 Hybrid Load Balancing and Traffic Management

Skill Badges



Google Cloud

[Build a Secure Google Cloud Network](#)



Google Cloud

[Implementing Cloud Load Balancing for Compute Engine](#)

Documentation

[Cloud Load Balancing documentation](#)

[Application Load Balancer overview](#)

[Cloud Load Balancing overview](#)

[Choosing a load balancer](#)

[Choosing a load balancer- traffic type](#)

[Load balancer feature comparison](#)

[External Application Load Balancer overview](#)

[Internal Application Load Balancer overview](#)

[Backend service-based external passthrough](#)

[Network Load Balancer overview](#)

[Internal passthrough Network Load Balancer overview](#)

[External proxy Network Load Balancer overview](#)

[Backend services overview](#)

[Forwarding rules overview](#)

[Instance groups](#)

[Creating managed instance groups](#)

[Network endpoint groups overview](#)

[Zonal network endpoint groups overview](#)

[Internet network endpoint groups overview](#)

[Serverless network endpoint groups overview](#)

[Proxy Network Load Balancer overview](#)

[Passthrough Network Load Balancer overview](#)

Let's consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you might want to include some or all of these.

[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Load Balancing \(On-demand\)](#)

[Implementing Cloud Load Balancing for Compute Engine \(Skill badge\)](#)

[Build a Secure Google Cloud Network \(Skill badge\)](#)

<https://docs.cloud.google.com/load-balancing/docs>

<https://docs.cloud.google.com/load-balancing/docs/application-load-balancer>

<https://cloud.google.com/load-balancing/docs/load-balancing-overview>

<https://cloud.google.com/load-balancing/docs/choosing-load-balancer>

<https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer#traffic-type>

[e](#)

<https://cloud.google.com/load-balancing/docs/features>

<https://cloud.google.com/load-balancing/docs/https>

<https://cloud.google.com/load-balancing/docs/il7-internal>

<https://cloud.google.com/load-balancing/docs/network/networklb-backend-service>
<https://cloud.google.com/load-balancing/docs/internal>
<https://cloud.google.com/load-balancing/docs/tcp>
<https://cloud.google.com/load-balancing/docs/backend-service>
<https://cloud.google.com/load-balancing/docs/forwarding-rule-concepts>
<https://cloud.google.com/compute/docs/instance-groups>
<https://cloud.google.com/compute/docs/instance-groups/creating-groups-of-managed-instances>
<https://cloud.google.com/load-balancing/docs/negs>
<https://cloud.google.com/load-balancing/docs/negs/zonal-neg-concepts>
<https://cloud.google.com/load-balancing/docs/negs/internet-neg-concepts>
<https://cloud.google.com/load-balancing/docs/negs/serverless-neg-concepts>
<https://cloud.google.com/kubernetes-engine/docs/how-to/horizontal-pod-autoscaling#autoscale-traffic>
<https://docs.cloud.google.com/load-balancing/docs/proxy-network-load-balancer>
<https://docs.cloud.google.com/load-balancing/docs/passthrough-network-load-balancer>

3.2 | Configuring Cloud CDN

Considerations include:

- Setting up Cloud CDN for supported origins (e.g., managed instance groups, Cloud Storage buckets, Cloud Run).
- Setting up Cloud CDN for external backends (Internet NEG) and third-party object storage.
- Invalidating cached content
- Configuring signed URLs

Google Cloud

Professional Cloud Network Engineers are expected to optimize network performance. A key aspect of this optimization is offloading the serving of static resources to edge locations to improve the availability, capacity, and performance and to reduce traffic on the network core. This capability is performed by content delivery networks (CDN) which have many features and configurations. Cloud CDN provides a managed CDN.

Questions 5, 6, and 7, tested your knowledge of Cloud CDN and its varying configuration options.

3.2 Diagnostic Question 05 Discussion



Cymbal Bank uses Cloud CDN to cache a web application served from a backend bucket connected to a Cloud Storage bucket. You need to cache all the web-app files with appropriate time to live (TTL) except for the index.html file. The index.html file contains links to versioned files and should always be fetched or re-validated from the origin.

Select the configuration option to satisfy these requirements with minimal effort.

- A. Set the Cloud CDN cache mode for the backend bucket to `CACHE_ALL_STATIC`.
- B. Set the Cloud CDN cache mode for the backend bucket to `FORCE_CACHE_ALL`, and ensure that the Cache-Control metadata for index.html is set to private.
- C. Set the Cloud CDN cache mode for the backend bucket to `CACHE_ALL_STATIC`, and ensure that the Cache-Control metadata for index.html is not set or is set to no-store, no-cache, or private.**
- D. Set the Cloud CDN cache mode to `USE_ORIGIN_HEADERS`, set the Cache-Control metadata for index.html to no-store, and set the Cache-Control headers for all the other files with appropriate TTL values.

Google Cloud

Feedback:

A. Incorrect. This option does not guarantee that the requirements are met. It would satisfy the requirements only if Cache-Control metadata on index.html is set appropriately.

B. Incorrect. This option would violate requirements and trigger caching for index.html

*C. Correct! This is the correct minimal effort configuration to ensure that requirements are met.

D. Incorrect. This option would ensure that the caching requirements are met, but would require setting or verifying Cache-Control metadata across all the files of the web-app.

Where to look:

<https://cloud.google.com/cdn/docs/overview>
<https://cloud.google.com/cdn/docs/features>
<https://cloud.google.com/cdn/docs/best-practices>
<https://cloud.google.com/cdn/docs/caching>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M12 Caching and Optimizing Load Balancing
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M2 Caching and Optimizing Load Balancing

Summary:

Cloud CDN has a simple configuration model that utilizes a cache mode parameter. The parameter can be one of three values.

1. `CACHE_ALL_STATIC` will cache static content based on Content-Type/MIME matching standard static types such as Javascript, CSS, photos, video, and audio, unless Cache-Control metadata for the associated object in Cloud Storage has a private or no-store directive. When the mode is `CACHE_ALL_STATIC`, a configuration parameter called Default TTL, which defaults to 1h, sets the lifetime for static file types that don't have Cache-Control specified expiry time. If they do have Cache-Control specified expiry time, they use the specified value. Other file types that are not in the set identified as static would be cached or not based on Cache-Control metadata normally.
2. `FORCE_CACHE_ALL` will enforce caching of all objects regardless of Cache-Control metadata and will also use the Default TTL value for expiry time.
3. `USE_ORIGIN_HEADERS` strictly uses the Cache-Control headers for controlling the caching. In addition to the Default TTL parameter there are also Max TTL and Client TTL configuration parameters that can adjust the behavior in the `CACHE_ALL_STATIC` and `FORCE_CACHE_ALL` modes.

3.2 Diagnostic Question 06 Discussion



Cymbal Bank is serving files from a backend bucket and wants to ensure time-limited read access without authentication. The backend bucket uses signed URLs to access those files. The files are also being cached in Cloud CDN. There is a problem with one of the files, and you want to delete it. You also want to immediately ensure no read access via the signed URL to the cached file copy in Cloud CDN, although the expiry time is currently set to sometime in the future.

Select the option that accomplishes this with lowest cost and effort.

- A. Perform cache invalidation for the file using the full path.
- B. Perform cache invalidation for the file using the path and excluding the query parameters used for the signed URL.
- C. Update the expiry time for the signed URL to be the current time.
- D. Delete the key used to create the signed URL.

Google Cloud

Feedback:

A. Incorrect. Cache invalidation requests would not consider query parameters and are not the most cost-effective approach.

B. Incorrect. Cache invalidation is not the lowest-cost approach to satisfy the requirements.

C. Incorrect. This action is not possible.

*D. Correct! Deleting the key will ensure that Cloud CDN rejects requests to the associated signed URLs and doesn't serve the cached values. This approach requires minimal cost and effort.

Where to look:

<https://cloud.google.com/cdn/docs/private-content>
<https://cloud.google.com/cdn/docs/using-signed-urls>
<https://cloud.google.com/cdn/docs/using-signed-cookies>
<https://cloud.google.com/cdn/docs/cache-invalidation-overview>
<https://cloud.google.com/cdn/docs/invalidating-cached-content>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M12 Caching and Optimizing Load Balancing
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M2 Caching and Optimizing Load Balancing

Summary:

Signed URLs and Cookies provide a mechanism to give controlled access to objects that are stored in Cloud Storage and cached in Cloud CDN.

3.2 Diagnostic Question 07 Discussion

Cymbal Bank provides premium market analysis reports in PDF format to paid subscribers via a global web portal. These files are stored in a private Cloud Storage bucket and served via Cloud CDN to minimize download latency. You need to ensure that only users with a valid active subscription can download these reports, without requiring them to have a Google Workspace account or authenticate directly against Google Cloud Identity. You want to minimize the load on your origin servers.



Select the configuration option which will meet the objective.

- A. **Configure Signed URLs on the Cloud CDN backend. Have your application generate a unique, time-bound URL for each authorized user request.**
- B. Configure Identity-Aware Proxy (IAP) on the Load Balancer backend service. Add the subscribers' email addresses to an access list with the IAP-secured Web App User role.
- C. Set up a VPC Service Control perimeter around the Cloud Storage bucket and the Load Balancer. Configure an ingress rule to allow traffic only from the subscribers' IP addresses.
- D. Create a Google Cloud Armor security policy with a custom rule that inspects the HTTP headers for a valid "Subscription-ID" token and attach it to the backend service.

Google Cloud

Feedback:

- *A. Correct. Signed URLs allow you to delegate authorization to your application, granting time-limited access to specific resources at the CDN edge without requiring Google credentials for end-users.
- B. Incorrect. Identity-Aware Proxy is designed for authenticating corporate users (employees/partners) with Google identities or OIDC, not for millions of external consumers accessing static assets.
- C. Incorrect. VPC Service Controls are used to prevent data exfiltration by defining network boundaries between Google Cloud services, not for authorizing individual user access to content.
- D. Incorrect. Google Cloud Armor is a Web Application Firewall (WAF) used to filter traffic based on security threats (DDoS, SQLi) or Geo-IP, not for cryptographic validation of individual user subscription rights.

Where to look:

<https://docs.cloud.google.com/cdn/docs/overview>
<https://cloud.google.com/cdn/docs/private-content>
<https://cloud.google.com/cdn/docs/using-signed-urls>
<https://cloud.google.com/cdn/docs/using-signed-cookies>
<https://cloud.google.com/cdn/docs/cache-invalidation-overview>
<https://cloud.google.com/cdn/docs/invalidating-cached-content>
<https://docs.cloud.google.com/cdn/docs/caching#cache-keys>
<https://docs.cloud.google.com/cdn/docs/using-cache-keys>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M12 Caching and Optimizing Load Balancing

Summary:

Signed URLs are the standard mechanism for serving private content via Cloud CDN to unauthenticated (public) users. In this architecture, your origin content (in Cloud Storage) remains private and is not directly accessible to the internet. When a user logs into your portal and requests a file, your application—which knows the user's subscription status—cryptographically signs the URL using a secret key shared with Cloud CDN.

This signed URL contains a signature and an expiration time. When the user's browser requests this URL, Cloud CDN validates the signature at the edge. If the signature is valid and the time has not expired, Cloud CDN serves the content from the cache. If the signature is invalid or expired, the request is rejected immediately at the edge, protecting your origin infrastructure from unauthorized traffic and ensuring that only paying users can access the cached content. This approach provides the least operational overhead for the networking team because authentication logic remains in the application layer, while delivery logic is offloaded to Google's global network. It avoids the complexity of managing Access Control Lists (ACLs) or IAM roles for thousands of external users, which would be unscalable and would negate the performance benefits of edge caching.

3.2 | Configuring Cloud CDN

Courses



[Networking in Google Cloud](#)

- M12 Caching and Optimizing Load Balancing



[Networking in Google Cloud: Load Balancing](#)

- M2 Caching and Optimizing Load Balancing

Documentation

[Cloud CDN overview](#)

[Cloud CDN features](#)

[Best practices for content delivery | Cloud CDN](#)

[Caching overview | Cloud CDN](#)

[Signed URLs and signed cookies overview | Cloud CDN](#)

[Using signed URLs | Cloud CDN](#)

[Using signed cookies | Cloud CDN](#)

[Cache invalidation overview | Cloud CDN](#)

[Invalidating cached content | Cloud CDN](#)

[Caching overview | cache-keys](#)

[Customize cache keys](#)

Let's consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Load Balancing \(On-demand\)](#)

<https://cloud.google.com/cdn/docs/overview>

<https://cloud.google.com/cdn/docs/features>

<https://cloud.google.com/cdn/docs/best-practices>

<https://cloud.google.com/cdn/docs/caching>

<https://cloud.google.com/cdn/docs/private-content>

<https://cloud.google.com/cdn/docs/using-signed-urls>

<https://cloud.google.com/cdn/docs/using-signed-cookies>

<https://cloud.google.com/cdn/docs/cache-invalidation-overview>

<https://cloud.google.com/cdn/docs/invalidating-cached-content>

3.3 | Configuring Cloud DNS

Considerations include:

- Managing Cloud DNS zones and records
- Migrating to Cloud DNS
- Configuring Cloud DNS routing policies such as geolocation and failover policies
- Enabling DNS Security Extensions (DNSSEC)
- Setting up self-hosted DNS integration with Cloud DNS including configuring DNS forwarding and DNS server policies
- Understanding DNS private and public zones and setting up-split-horizon DNS
- Setting up DNS cross-project binding and DNS peering
- Configuring Cloud DNS and external-DNS operator for GKE

Google Cloud

A Professional Cloud Network Engineer is responsible for designing and managing the private and public DNS of their organization.

Question 8 explored Cloud DNS managed zones and DNS record configuration.

Question 9 tested your ability to configure Cloud DNS peering and forwarding.

Question 10 verified your understanding of DNSSEC (Domain Name System Security Extensions).

3.3 Diagnostic Question 08 Discussion



Cymbal Bank will use a hybrid DNS approach. Cymbal has a VPC in Google Cloud that connects to their on-premises networks via Interconnect. You will use Cloud DNS for Cymbal's public DNS zone at cymbalbank.com and also for private DNS for resources at gcp.cymbalbank.com. You will use Cymbal's on-premises DNS, which is configured as authoritative for on-premises private resources at corp.cymbalbank.com.

Select the Cloud DNS managed zone configuration that will satisfy the requirements.

- A. Create a single Cloud DNS managed zone in Google Cloud that is configured for private DNS for gcp.cymbalbank.com and public DNS for cymbalbank.com and that also acts as a forwarding zone to the on-premises DNS for corp.cymbalbank.com DNS requests.
- B. **Create a Cloud DNS private managed zone for gcp.cymbalbank.com, a public managed zone for cymbalbank.com, and a third forwarding zone for corp.cymbalbank.com that forwards DNS requests to the on-premises DNS.**
- C. Create a public managed zone for cymbalbank.com and a Cloud DNS private managed zone for gcp.cymbalbank.com that also forwards DNS requests for corp.cymbalbank.com to the on-premises DNS.
- D. Create a Cloud DNS private managed zone for gcp.cymbalbank.com and a public managed zone for cymbalbank.com that also forwards DNS requests for corp.cymbalbank.com to the on-premises DNS.

Google Cloud

Feedback:

- A. Incorrect. This configuration is invalid. A single managed zone cannot be both public and private. Also forwarding zones must also be a separate managed zone.
- *B. Correct! This is the valid configuration option.
- C. Incorrect. This configuration is invalid. A separate forwarding zone is necessary to send DNS requests to the on-premises DNS.
- D. Incorrect. This configuration is invalid. A separate forwarding zone is necessary to send DNS requests to the on-premises DNS.

Where to look:

<https://cloud.google.com/dns/docs/overview>
<https://cloud.google.com/dns/docs/dns-overview>
<https://cloud.google.com/dns/docs/best-practices>
<https://cloud.google.com/dns/docs/key-terms>
<https://cloud.google.com/dns/docs/zones>
<https://cloud.google.com/dns/docs/records>
<https://cloud.google.com/dns/docs/dnssec>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M4 Network Routing and Addressing
- On-demand course: **Networking in Google Cloud: Routing and Addressing**
 - M1 Network Routing and Addressing

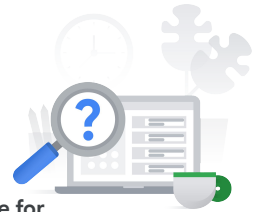
Summary:

Cloud DNS allows for creation of managed zones that model DNS zones. These zones can contain DNS record sets for authoritative DNS serving for a specified public or private domain.

Cloud DNS also supports many other capabilities including DNS forwarding and peering, but requires valid configuration to support these DNS scenarios. Separate managed zones must be created for different domains, public vs private DNS, DNS peering, or DNS forwarding.

Cloud DNS also allows for DNS server policies to be created for VPCs that allow inbound or outbound forwarding of DNS requests. For outbound DNS forwarding, both outbound DNS server policies and forwarding zones can be used. When forwarding DNS requests to on-premises DNS servers, forwarding zones are the recommended approach.

3.3 Diagnostic Question 09 Discussion



You are configuring hybrid DNS for Google Cloud using Cloud DNS and your on-premises DNS. You have three VPC networks in Google Cloud in three different projects that should forward DNS requests for a particular private domain to the on-premises DNS. All three projects have Cloud VPN connections to the on-premises network.

Select the Google recommended approach for enabling this requirement.

- A. For the VPC in one of the projects, create a Cloud DNS forwarding zone for its VPC. For the VPC in each of the other projects, create a Cloud DNS peering zone that targets the VPC with the forwarding zone.
- B. Create a forwarding zone in one of the projects that is visible to the VPCs in all of the projects.
- C. Create a forwarding zone in each of the projects that is visible to the VPC in that project.
- D. Create a forwarding zone and a peering zone in each project. Make the forwarding zone visible to the VPC in the same project and the peering managed zones associated with the VPCs in the other projects.

Google Cloud

Feedback:

*A. Correct! This approach ensures that all DNS requests and responses are routed correctly to and from the on-premises network. It also satisfies the visibility requirements: private managed zones and forwarding zones are only visible to VPCs in the same project.

B. Incorrect. This approach is not possible. A forwarding zone can only be visible to VPCs in the same project.

C. Incorrect. This approach is not recommended. It would not function properly if all the VPCs were connected to the same on-premises network for routing reasons. The Cloud Routers would all advertise different routes for the same destination IP range in Google Cloud for the responses to Cloud DNS requests.

D. Incorrect. This approach is not possible. A peering managed zone can only be peered with one other project and VPC and also has visibility to one or more VPCs in the local project.

Where to look:

<https://cloud.google.com/dns/docs/vpc-name-res-order>

<https://cloud.google.com/dns/docs/policies-overview>

<https://cloud.google.com/dns/docs/zones/cross-project-binding>

<https://cloud.google.com/dns/docs/server-policies-overview>

<https://cloud.google.com/dns/docs/zones/manage-response-policies>

<https://cloud.google.com/dns/docs/zones/manage-routing-policies>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M4 Network Routing and Addressing
- On-demand course: **Networking in Google Cloud: Routing and Addressing**
 - M1 Network Routing and Addressing

Summary:

DNS Peering is recommended to avoid outbound DNS forwarding from multiple VPCs which can cause problems with return traffic. DNS peering allows a single forwarding zone to be associated with a single VPC and then other VPCs to have their requests forwarded by DNS peering with the forwarding zone.

3.3 Diagnostic Question 10 Discussion



Cymbal Bank hosts its public marketing site `cymbalbank.com` using Cloud DNS. The security team has mandated that the domain must be protected against cache poisoning and DNS spoofing attacks to ensure customers are not redirected to malicious sites.

- A. **Enable DNSSEC on the Cloud DNS managed zone and copy the DS record to the domain registrar.**
- B. Create a Cloud DNS Response Policy to block malicious IPs.
- C. Configure a private managed zone for `cymbalbank.com` and use DNS Peering.
- D. Enable Google Cloud Armor on the Load Balancer associated with the domain.

Select the steps you must take to secure the zone?

Google Cloud

Feedback:

- *A. Correct! DNSSEC (Domain Name System Security Extensions) authenticates DNS responses using cryptographic signatures. This prevents attackers from manipulating or poisoning the DNS cache to redirect users to fake websites.
- B. Incorrect. Response policies are used for "firewalling" DNS (blocking resolution of known bad domains) for outbound lookups from your VPC, not protecting your own domain from spoofing.
- C. Incorrect. The scenario specifies a public marketing site. Private zones are not resolvable from the public internet.
- D. Incorrect. Google Cloud Armor protects the Application Load Balancer (Layer 7) from DDoS and web attacks, but it does not protect the DNS lookup process itself.

Where to look:

<https://cloud.google.com/dns/docs/overview>
<https://cloud.google.com/dns/docs/dns-overview>
<https://cloud.google.com/dns/docs/best-practices>
<https://cloud.google.com/dns/docs/key-terms>
<https://cloud.google.com/dns/docs/zones>
<https://cloud.google.com/dns/docs/records>
<https://cloud.google.com/dns/docs/dnssec>

Content mapping:

- ILT course: **Networking in Google Cloud**

- M4 Network Routing and Addressing
- On-demand course: **Networking in Google Cloud: Routing and Addressing**
 - M1 Network Routing and Addressing

Summary:

DNSSEC (Domain Name System Security Extensions) authenticates DNS responses using cryptographic signatures. This prevents attackers from manipulating or poisoning the DNS cache to redirect users to fake websites. To implement this in Google Cloud, you must first enable DNSSEC on the Managed Zone.

Once enabled, Cloud DNS generates specific DNS records (DNSKEY and DS). You must then take the DS (Delegation Signer) record provided by Cloud DNS and register it with your domain registrar (where you bought the domain). This establishes the "chain of trust" required for valid DNSSEC resolution.

3.3 | Configuring Cloud DNS

Courses



[Networking in Google Cloud](#)

- M4 Network Routing and Addressing



[Networking in Google Cloud: Routing and Addressing](#)

- M1 Network Routing and Addressing

Documentation

[Cloud DNS overview](#)

[General DNS overview](#)

[DNS best practices](#)

[Key terms | Cloud DNS](#)

[Manage zones | Cloud DNS](#)

[Manage records | Cloud DNS](#)

[DNS Security Extensions \(DNSSEC\) overview](#)

[Name resolution order | Cloud DNS](#)

[DNS policies overview](#)

[Cross-project binding zones | Cloud DNS](#)

[DNS server policies](#)

[Manage response policies and rules | Cloud DNS](#)

[Manage DNS routing policies](#)

Let's consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these links.

[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Routing and Addressing \(On-demand\)](#)

<https://cloud.google.com/dns/docs/overview>

<https://cloud.google.com/dns/docs/dns-overview>

<https://cloud.google.com/dns/docs/best-practices>

<https://cloud.google.com/dns/docs/key-terms>

<https://cloud.google.com/dns/docs/zones>

<https://cloud.google.com/dns/docs/records>

<https://cloud.google.com/dns/docs/dnssec>

<https://cloud.google.com/dns/docs/vpc-name-res-order>

<https://cloud.google.com/dns/docs/policies-overview>

<https://cloud.google.com/dns/docs/zones/cross-project-binding>

<https://cloud.google.com/dns/docs/server-policies-overview>

<https://cloud.google.com/dns/docs/zones/manage-response-policies>

<https://cloud.google.com/dns/docs/zones/manage-routing-policies>